1.1 Evaluating Algebraic Expressions

Name _____ Period ____ Date _____

Algebra I 1.1 (30-minute plan) · Student sheet

I can…

- evaluate expressions with one variable.
- evaluate expressions with two or more variables using the order of operations.
- write and evaluate expressions for real-life situations.

Vocabulary · fill in the missing word

Variable — a _____ that stands for a number

Algebraic expression — an expression with numbers, variables, and _____

Evaluate — _____ numbers for the variables and simplify

Order of operations — grouping symbols, _____, multiply and divide (left to right), add and subtract (left to right)

Part 1 · Notes and examples (fill in each note, then work the example)

Order of operations: Order of operations: _____, _____, multiply and _____ (left to right), add and _____ (left to right).

Evaluating: To evaluate, _____ each variable with its value, using _____ around negatives.

Careful with negatives: -3^2 = _____, but $(-3)^2$ = _____.

Example 1

Evaluate each expression.

a. $5x - 8$ when $x = 4$ b. $2x^2 + 1$ when $x = -3$

Example 2

A phone plan costs $25 a month plus $0.10 per text. The cost is $C = 25 + 0.1t$, where t is the number of texts. Find the cost for 140 texts.

Part 2 · Partner practice

Partner practice

Evaluate $3x^2 - x$ when $x = -2$.